Aakash
Medical | IIT-JEE | Foundations
*Corporate Office :* AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, ***Ph.***+91-1244168300

**MM : 720** **NCERT Booster Test Series for NEET-2026_RM(P4)_NBTS-03D** **Time : 180 Min.**

**Topics Covered:**
**Physics:** Alternating Current, Electromagnetic Waves, Ray Optics & Optical Instruments, Wave Optics, Dual Nature of Radiation and Matter, Atoms, Nuclei, Semiconductor, Experimental Skills
**Chemistry:** Amines (Organic Compound containing Nitrogen), The d & f-Block Elements, Coordination Compounds, The p-Block Elements, Biomolecules, Principles Related to Practical Chemistry
**Botany:** Molecular Basis of Inheritance, Microbes in Human Welfare, Organisms and Populations, Ecosystem, Biodiversity and Conservation
**Zoology:** Evolution, Human Health & Diseases, Biotechnology: Principles & Processes, Biotechnology & Its Applications

**General Instructions :**
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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## PHYSICS

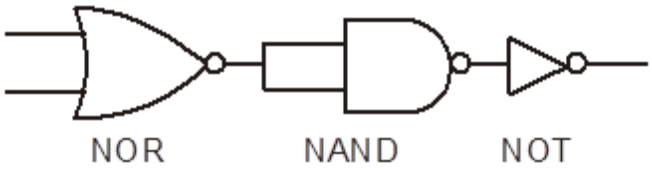
**1.** The de-Broglie's wavelength associated with a particle of mass $m$ = 2 kg and moving with a velocity 100 m/s is
(1) $6.6 \times 10^{-34}$ m (2) $3.3 \times 10^{-36}$ m
(3) $3.3 \times 10^{-34}$ m (4) $5 \times 10^{-37}$ m

**2.** Given below are two statements:
**Statement I :** A time varying magnetic field can produce a time varying electric field.
**Statement II :** An electric charge oscillating harmonically with frequency $f$, produces electromagnetic waves of the same frequency $f$.
In the light of the above statements, choose the most appropriate answer from the options given below.
(1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect
(3) Statement I is correct but Statement II is incorrect
(4) Statement I is incorrect but statement II is correct

**3.** A wire which follows Hooke's law extends by 1.5 cm when a certain force is applied. If double the force is applied to another wire of same material and length, but twice the radius of previous one then elongation produced in another wire is
(1) 8 mm (2) 0.9 cm
(3) 0.75 cm (4) 3 cm

**4.** Choose the correct Alternative
**Statement-A:** Number of protons and number of neutrons are conserved in each nuclear reaction.
**Statement-B:** Conservation of energy takes place only in nuclear reaction and never in chemical reaction.
(1) Only A is correct (2) Only B is correct
(3) Both A & B are correct (4) Both A & B are incorrect

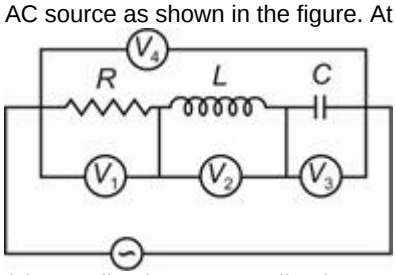
**5.** The logic circuit given below is equivalent to




(1) OR gate (2) NOR gate
(3) AND gate (4) NAND gate

**6.** Refractive index of a material is equal to tangent of polarizing angle. It is called
(1) Brewster's law (2) Lambert's law
(3) Malus's law (4) Bragg's law

**7.** An ideal resistance *R*, ideal inductance *L*, ideal capacitance *C* and *AC* voltmeters $V_1$, $V_2$, $V_3$ and $V_4$ are connected to an AC source as shown in the figure. At resonance




(1) Reading in $V_3$ = Reading in $V_1$
(2) Reading in $V_1$ = Reading in $V_2$
(3) Reading in $V_2$ = Reading in $V_4$
(4) Reading in $V_2$ = Reading in $V_3$

**8.** Suppose a pure Ge crystal has $5 \times 10^{34}$ atoms per metre cube. It is doped by 1 ppm concentration of pentavalent atom. The hole concentration of doped semiconductor will be ($n_i = 2.5 \times 10^{19}$ m$^{-3}$)

(1) $2.5 \times 10^{9}$ m$^{-3}$ (2) $5 \times 10^{10}$ m$^{-3}$
(3) $1.25 \times 10^{8}$ m$^{-3}$ (4) $1.25 \times 10^{10}$ m$^{-3}$

**9.** A : Diffraction of sound is commonly observed in everyday life.
R : Diffraction of light is easily possible.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

**10.** In the $n^{th}$ state, energy of electron in hydrogen atom is $E_n = -\frac{13.6}{n^2}$ eV. The potential energy of electron in first excited state will be

(1) –27.2 eV (2) –3.4 eV
(3) –3.2 eV (4) –6.8 eV

**11.** A photoelectric surface is illuminated successively by monochromatic light of wavelength λ and $\frac{\lambda}{2}$. If the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case, the work function of the surface of the material is
(*h* = Planck's constant, *c* = speed of light)

(1) $\frac{2hc}{\lambda}$ (2) $\frac{hc}{2\lambda}$
(3) $\frac{hc}{3\lambda}$ (4) $\frac{hc}{\lambda}$

**12.** A nucleus of mass number A has mass defect $\Delta m$. What is the formula for the binding energy per nucleon of this nucleus? (*C* = velocity of light)

(1) $\frac{\Delta mc^2}{A}$ (2) $\frac{\Delta m^2 c}{A}$
(3) $\frac{\Delta mc}{A}$ (4) $(\Delta mc^2)A$

**13.** In a extrinsic semiconductor, the impurity level is just below conduction band. The semiconductor is

(1) p-type (2) n-type
(3) Intrinsic (4) May be p-type or n-type

**14.** If series *LCR* circuit is connected across an a.c. source $E = 20\sin(50\pi t)$ V. Current from the supply is $I = 1\sin\left(50\pi t + \frac{\pi}{3}\right)$ A. The average power dissipated per cycle is

(1) 5 W (2) 2 W
(3) 10 W (4) 8 W

**15.** The de-Broglie wavelength of an atom of mass *m* at absolute temperature *T* will be (symbols have their usual meaning)

(1) $\frac{h}{\sqrt{2mKT}}$ (2) $\frac{h}{\sqrt{mKT}}$
(3) $\frac{h}{\sqrt{3mKT}}$ (4) $\sqrt{2mKT}$

**16.** A : Electromagnetic waves have momentum and energy.
R : Electromagnetic waves do not need any material medium for its propagation.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

**17.** A capillary tube of radius '*r*' is immersed in a liquid of surface tension (*T*). If angle of contact is 0°, then mass of liquid which rises up in capillary is

(1) $\frac{T}{2g}(\pi r)$ (2) $\frac{2T}{g}(2\pi r)$
(3) $\frac{T}{g}(2\pi r)$ (4) $\frac{T}{g}(\pi r)$

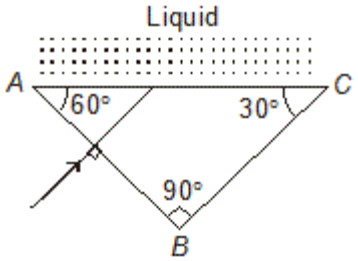
**18.** 1 a.m.u is equal to

(1) $\frac{1}{20}$ (mass of one O-atom) (2) $\frac{1}{12}$ (mass of one C-atom)
(3) $\frac{1}{18}$ (mass of one N-atom) (4) $\frac{1}{24}$ (mass of one S-atom)

**19.** Wave theory cannot explain the phenomenon of
(A) Polarisation
(B) Compton effect
(C) Diffraction
(D) Photoelectric effect

(1) (A) and (C) (2) (B) and (C)
(3) (B) and (D) (4) (A) and (D)

**20.** Light is incident normally on face *AB* of a prism as shown in figure. A liquid of refractive index μ is placed on face *AC* of the prism, the prism is made of glass of refractive index $\frac{3}{2}$. The limit of μ for which total internal reflection takes place on face *AC* is




(1) μ > $\frac{3}{4}$ (2) μ < $\frac{3\sqrt{3}}{4}$
(3) μ > $\sqrt{3}$ (4) μ < $\frac{7\sqrt{3}}{4}$

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**21.** In YDSE experiment, electron beam is used in place of light which was accelerated through some potential difference. If accelerating potential difference is made four times, then width of fringes will become
(1) 2 times (2) 4 times
(3) $\frac{1}{2}$ times (4) $\frac{1}{4}$ times

**22.** According to Bohr's model for hydrogen and hydrogen type atoms. Match the Column I with Column II.

| | Column I | | Column II |
|---|---|---|---|
| (a) | Frequency | (i) | Proportional to n |
| (b) | Angular momentum | (ii) | Proportional to $\frac{Z}{n}$ |
| (c) | Linear momentum | (iii) | Proportional to $\frac{Z^2}{n^3}$ |
| (d) | Radius | (iv) | Proportional to $\frac{n^2}{Z}$ |

(1) (a)(i); (b)(iii); (c)(ii); (d)(iv) (2) (a)(iii); (b)(i); (c)(iv); (d)(ii)
(3) (a)(ii); (b)(i); (c)(iii); (d)(iv) (4) (a)(iii); (b)(i); (c)(ii); (d)(iv)

**23.** A radiation of energy *E* falls normally on a perfectly absorbing surface. The momentum transferred to the surface is (*c* = velocity of light)
(1) $\frac{E}{c^2}$ (2) $\frac{2E}{c}$
(3) $\frac{2E}{c^2}$ (4) $\frac{E}{c}$

**24.** A plane electromagnetic wave of frequency 30 Hz travels in free space along *x* direction. At a particular point of space and time $\vec{E} = 7.2\hat{j}\ \mathrm{V/m}$, then $\vec{B}$at this point is
(1) $10.2 \times 10^{-8} T\hat{k}$ (2) $21.6 \times 10^{-8}\mathrm{Tk}$
(3) $24 \times 10^{-9} T\hat{k}$ (4) $21.6 \times 10^{8} T\hat{k}$

**25.** What is the respective number of α and $\beta^-$ particles, emitted in the following radioactive decay?
${}_{90}X^{200} \rightarrow {}_{80}Y^{168}$
(1) 6 and 8 (2) 8 and 8
(3) 6 and 6 (4) 8 and 6

**26.** Boolean expression $\mathrm{Y} = \overline{\overline{\mathrm{A}} + \overline{\mathrm{B}}}$, can also be written as
(1) $\mathrm{A} \cdot \mathrm{B}$ (2) $\overline{\mathrm{A}} \cdot \mathrm{B}$
(3) $\overline{\mathrm{A}} + \overline{\mathrm{B}}$ (4) $\overline{\mathrm{A} \cdot \mathrm{B}}$

**27.** The radius of second stationary orbit of electrons in Bohr's atom is *R*. The radius of the third orbit will be
(1) 3*R* (2) 2.25*R*
(3) 9*R* (4) $\frac{R}{3}$

**28.** Ratio of shortest wavelengths corresponding to Lyman and Balmer series in hydrogen spectrum is
(1) $\frac{1}{4}$ (2) $\frac{1}{9}$
(3) $\frac{5}{27}$ (4) $\frac{16}{27}$

**29.** A compound microscope has a magnifying power 30. Focal length of its eye piece is 5 cm. If the final image formed at least distance of distinct vision then magnification produced by the objective is
(1) 2 (2) 6
(3) 5 (4) 3

**30.** In an experiment of photoelectric effect, slope of stopping potential versus frequency of incident light will be (Symbols have their usual meaning)
(1) *h*/*e* (2) *he*
(3) *e* (4) *h*

**31.** Instantaneous value of electric current varies in a wire according to $i = [8 + 15\sin(200\ \pi t)]$ ampere. Its average value over one complete cycle is given as
(1) 6 A (2) 8 A
(3) 23 A (4) 2.3 A

**32.** A prism has a refracting angle of 75°. When placed in the position of minimum deviation, it produces a deviation of 45°, the angle of incidence is
(1) 30° (2) 45°
(3) 60° (4) 90°

**33.** **Statement A:** For a cosine wave input voltage, ratio of average value of ac to virtual value of ac is 2
**Statement B:** Average value of ac and virtual value of ac are equal
Choose the correct option
(1) Both A and B are true (2) A is true but B is wrong
(3) B is true but A is wrong (4) Both A and B are wrong

**34.** When unpolarized beam of light of intensity $I_0$ is incident on two polarizer in contact. If the angle between the axes of two polarizers is 60°, then the intensity of light finally emerging from the combination is
(1) $\frac{I_0}{8}$ (2) $\frac{I_0}{4}$
(3) $\frac{3I_0}{8}$ (4) $\frac{I_0}{2}$

**35.** A fish at a depth of 12 cm in water is viewed by an observer on the bank of a lake. To what height, the image of the fish is raised? (Take $\mu_{\text{water}} = \frac{4}{3}$)
(1) 9 cm (2) 12 cm
(3) 3.8 cm (4) 3 cm

**36.** In a resonance tube experiment a closed organ pipe of length 100 cm is used. Initially it is completely filled with water. It is vibrated with tuning fork of frequency 340 Hz. To achieve resonance the water level is lowered then (given $v_{\text{air}}$ = 340 m/s, neglect end correction)
(1) First resonating length is 30 cm
(2) First resonating length is 25 cm
(3) Second resonating length is 75 cm
(4) Both (2) & (3)

**37.** The binding energy per nucleon for ${}_1H^1$, ${}_1H^3$ and ${}_2He^4$ are *a*, *b* and *c* respectively. Then energy released in a hypothetical nuclear reaction ${}_1H^1 + {}_1H^3 \rightarrow {}_2He^4$ + energy, is
(1) *a* + 3*b* – 4*c* (2) 4*c* – *a* – 3*b*
(3) 4*c* – *a* + 3*b* (4) *c* – *a* – *b*

**38.** Two convex lens $L_1$ and $L_2$ of focal length 15 cm and 10 cm respectively are placed at *d* distance apart. If rays parallel to principal axis fall on lens $L_1$ and these rays emerge parallel to principal axis after passing through lens $L_2$ then the value of *d* is
(1) 5 cm (2) 10 cm
(3) 15 cm (4) 25 cm

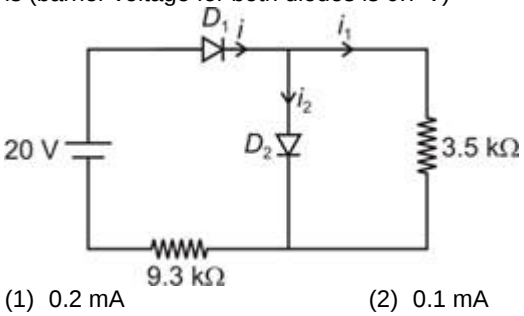
**39.** In hydrogen atom, for which transition maximum energy is emitted?
(1) $n_1 = 2, n_2 = 3$ (2) $n_1 = 2, n_2 = 4$
(3) $n_1 = 3, n_2 = 4$ (4) $n_1 = 1, n_2 = 3$

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**40.** Originally the Young's double slit experiment was performed in white light in which higher order spectra in the fringe pattern overlap each other. Assume that wavelength of blue light is 520 nm and that of red light is 780 nm. The value of $x$ for which $x^{th}$ bright red band coincides with $(x + 1)^{th}$ blue band is

(1) 5 (2) 4
(3) 3 (4) 2

**41.** In the circuit shown in figure, current through 3.5 kΩ resistor is (barrier voltage for both diodes is 0.7 V)




(1) 0.2 mA (2) 0.1 mA
(3) 1.2 mA (4) 1.8 mA

**42.** Power factor of series LCR circuit at resonance is

(1) $\frac{1}{\sqrt{2}}$ (2) $\frac{\sqrt{3}}{2}$
(3) 1 (4) Zero

**43.** An object is placed at a distance of 80 cm from a screen. Where should a convex lens of focal length 15 cm be placed from object such that a real image of object is formed?

(1) 60 cm (2) 20 cm
(3) 40 cm (4) Both (1) and (2)

**44.** In a Vernier callipers 19 divisions of main scale exactly coincide with 20 divisions of Vernier scale. If smallest division of main scale is 1 mm, what is Vernier constant of callipers?

(1) 0.05 mm (2) 0.005 mm
(3) 0.01 mm (4) 0.001 mm

**45.** Which of the following statements is correct about the electromagnetic wave?

(1) Both electric and magnetic field vectors are in opposite phase
(2) Electromagnetic wave always requires a medium for propagation
(3) The energy in the electromagnetic wave is divided equally between electric and magnetic fields
(4) Both (2) and (3)

## CHEMISTRY

**46.** Match **List-I** with **List-II**

| List-I | List-II |
|---|---|
| a. Epsom salt | (i) $CuFeS_2$ |
| b. Baryte | (ii) $MgSO_4 \cdot 7H_2O$ |
| c. Copper pyrites | (iii) $CaSO_4 \cdot 2H_2O$ |
| d. Gypsum | (iv) $BaSO_4$ |

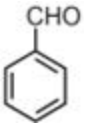
Choose the correct option.

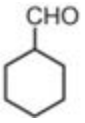
(1) a(i), b(ii), c(iii), d(iv) (2) a(ii), b(iii), c(i), d(iv)
(3) a(ii), b(iv), c(i), d(iii) (4) a(iv), b(i), c(ii), d(iii)

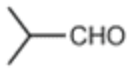
**47.** The correct order of boiling point of hydrides of group 15 elements is

(1) $NH_3 > PH_3 > AsH_3 > SbH_3$
(2) $SbH_3 > AsH_3 > PH_3 > NH_3$
(3) $SbH_3 > NH_3 > AsH_3 > PH_3$
(4) $NH_3 > SbH_3 > PH_3 > AsH_3$

**48.** Which among the following compounds does not give positive Fehling test?

(1) 


(2) 


(3) 


(4) HCHO

**49.** Oxidation of glucose with bromine water and nitric acid yields respectively

(1) Gluconic acid and saccharic acid
(2) Saccharic acid and gluconic acid
(3) Gluconic acid and gluconic acid
(4) Saccharic acid and saccharic acid

**50.** The non-reducing sugar among the following is

(1) Galactose (2) Maltose
(3) Lactose (4) Sucrose

**51.** Which of the following contain N – N bond?

(a) $N_2O_3$
(b) $N_2O_5$
(c) $N_2O_4$
(d) $N_2O$

Select the correct option

(1) (a), (b) and (c) only (2) (b), (c) and (d) only
(3) (a), (c) and (d) only (4) (a), (b), (c) and (d)

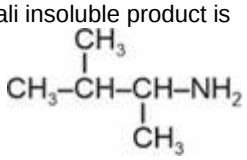
**52.** Consider the following statements about colloids

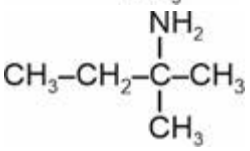
(a) Generally, size of colloidal particles ranges from 1–1000 nm
(b) Starch sol is an example of lyophilic sol.
(c) Stability of lyophobic sol is due to charge on colloidal particles.

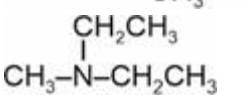
The correct statements are

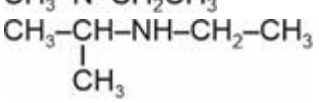
(1) (a) and (b) only (2) (b) and (c) only
(3) (a) and (c) only (4) (a), (b) and (c)

**53.** The amine that reacts with Hinsberg's reagent to give an alkali insoluble product is

(1) 


(2) 


(3) 


(4) 


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**54.** For titration of weak acid and strong base the suitable indicator is
(1) Methyl orange (2) Methyl red
(3) Phenolphthalein (4) Litmus

**55.** The complex which will absorb shortest wavelength of light is
(1) $[Ni(CN)_6]^{4-}$ (2) $[Ni(NH_3)_6]^{2+}$
(3) $[Ni(H_2O)_6]^{2+}$ (4) $[NiCl_4]^{2-}$

**56.** Which of the following element does not show the variable oxidation state?
(1) Fe (2) Mn
(3) Cu (4) Zn

**57.** Match List-I with List-II for structure of Borax.

| List-I | List-II |
|---|---|
| (a) Number of hydrated water molecules | (i) 2 |
| (b) Number of B–O–B bonds | (ii) 8 |
| (c) Number of $sp^2$ boron | (iii) 5 |
| (d) Number of B–OH bonds | (iv) 4 |

Correct match is
(1) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)
(2) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
(3) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
(4) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

**58.** Match the cations present in the **List-I** with their group reagent given in **List-II.**

| List-I | List-II |
|---|---|
| a. $Al^{3+}$ | (i) $H_2S$ in presence of $NH_4OH$ |
| b. $Ni^{2+}$ | (ii) $(NH_4)_2CO_3$ in presence of $NH_4OH$ |
| c. $Ba^{2+}$ | (iii) $H_2S$ gas in presence of dil HCl |
| d. $Cu^{2+}$ | (iv) $NH_4OH$ in presence of $NH_4Cl$ |

The correct match is
(1) a(i), b(iv), c(iii), d(ii) (2) a(iv), b(ii), c(i), d(iii)
(3) a(ii), b(iii), c(i), d(iv) (4) a(iv), b(i), c(ii), d(iii)

**59.** How many moles of NaOH are required to react completely with 4 moles of chlorine gas, when it is passed in hot and concentrated NaOH solution?
(1) 6 (2) 5
(3) 10 (4) 8

**60.** Glucose on prolonged heating with HI gives
(1) Gluconic acid (2) n-Hexane
(3) Saccharic acid (4) Hexanoic acid

**61.** Consider the following statements about interstitial compounds
(i) They are chemically inert
(ii) They are very hard
(iii) They retain metallic conductivity
The correct statements are
(1) (i) and (ii) only (2) (ii) and (iii) only
(3) (i) and (iii) only (4) (i), (ii) and (iii)

**62.** In aqueous solution, the correct order of basic strength of amines is
(1) $CH_3NH_2 > (CH_3)_2NH > (CH_3)_3N$
(2) $(CH_3)_3N > (CH_3)_2NH > CH_3NH_2$
(3) $(CH_3)_2NH > (CH_3)_3N > CH_3NH_2$
(4) $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N$

**63.** Number of moles of $KMnO_4$ required to react completely with 1 mole of $FeSO_4$ in acidic medium is
(1) 5 (2) $\frac{1}{5}$
(3) 3 (4) $\frac{1}{3}$

**64.** Given below are two statements
**Statement I:** On fusion of $FeCr_2O_4$ with potassium carbonate in free access of air yields potassium chromate.
**Statement II:** Sodium dichromate is more soluble than potassium dichromate.
In the light of above statements choose the correct option
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

**65.** Match the processes given in List-I with their catalysts given in List-II

| | List-I | | List-II |
|---|---|---|---|
| (a) | Haber's process | (i) | $V_2O_5$ |
| (b) | Ostwald's process | (ii) | Iron oxide with small amounts of $K_2O$ and $Al_2O_3$ |
| (c) | Contact process | (iii) | $CuCl_2$ |
| (d) | Deacon's process | (iv) | Pt/Rh gauge |

Select the **correct** option.
(1) a(i), b(ii), c(iv), d(iii) (2) a(iv), b(ii), c(i), d(iii)
(3) a(iii), b(iv), c(ii), d(i) (4) a(ii), b(iv), c(i), d(iii)

**66.** Which of the following ions impart brick red colour to the flame test (observed by naked eye)?
(1) $Mg^{2+}$ (2) $Ca^{2+}$
(3) $Sr^{2+}$ (4) $Ba^{2+}$

**67.** Number of chiral carbons in cyclic structure of α-D-fructose is
(1) 3 (2) 4
(3) 5 (4) 6

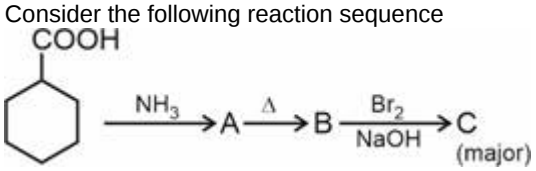
**68.** Choose the correct statement from the following.
(1) Sucrose is dextrorotatory but after hydrolysis gives laevorotatory glucose and dextrorotatory fructose
(2) Maltose is composed of one α-D-Glucose unit and one β-D-Fructose
(3) Lactose is composed of one β-D-Galactose and one β-D-Glucose units
(4) Amylose is water insoluble component which constitutes about 80-85% starch

**69.** Given below are two statements.
**Statement I:** Geometrical isomerism is not possible in tetrahedral complexes.
**Statement II:** $[MA_3B_3]$ type complexes show geometrical isomerism.
In the light of above statements choose the correct option given below.
(1) Both statements I and II are correct
(2) Both statements I and II are incorrect
(3) Statement I is incorrect but statement II is correct.
(4) Statement I is correct but statement II is incorrect

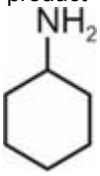
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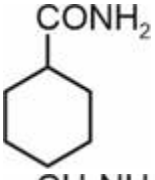
**70.** Given below are two statements: one is labelled as Assertion (A) and other is labelled as a Reason (R)
**Assertion (A):** Manganate ion is paramagnetic in nature.
**Reason (R):** Manganate ion contains one unpaired electron.
In light of above statements, choose the correct answer.
(1) Both (A) and (R) are true statements and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true statements but (R) is not the correct explanation of (A)
(3) (A) is false but (R) is true
(4) (A) is true but (R) is false

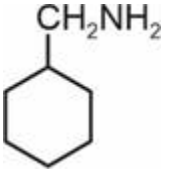
**71.** Consider the following reaction sequence

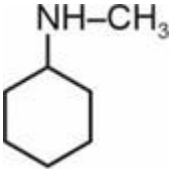



The product 'C' is

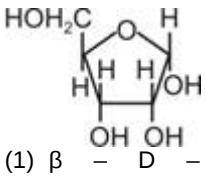
(1) 


(2) 


(3) 

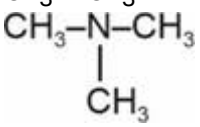

(4) 

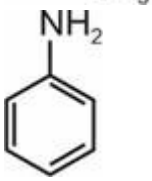

**72.** One mole of which octahedral complex of cobalt will give 2 moles of AgCl as precipitate when reacts with excess of silver nitrate solution?
(1) $CoCl_3 \cdot 6NH_3$ (2) $CoCl_3 \cdot 4NH_3$
(3) $CoCl_3 \cdot 5NH_3$ (4) $CoCl_3 \cdot 3NH_3$

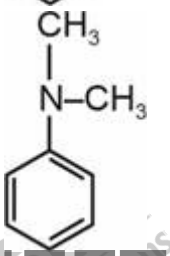
**73.**  is known as
(1) β – D – 2 – deoxyribose (2) α – D – 2 – deoxyribose
(3) β – D – ribose (4) α – D – ribose

**74.** On addition of dilute $H_2SO_4$ in a salt solution, a colourless and odourless gas is evolved with brisk effervescence which turns lime water milky. Anion present in the salt may be
(1) $NO_3^-$ (2) $Cl^-$
(3) $NO_2^-$ (4) $CO_3^{2-}$

**75.** Consider the following statements
(I) Zr and Hf have almost same ionic radii
(II) Lanthanoids liberate $H_2$ gas when treated with dilute acids.
(III) The ionic radii of trivalent lanthanoids steadily increases with increase in atomic number.
The correct statement(s) is/are
(1) (I) and (II) only (2) (I) and (III) only
(3) Only (II) (4) (I), (II) and (III)

**76.** Which of the following amines will give the positive carbylamine test?
(1) $CH_3NHCH_3$
(2) 

(3) 

(4) 


**77.** The hybridization and shape of the complex $[Ni(CO)_4]$ are
(1) $sp^3$, square planar (2) $sp^3$, tetrahedral
(3) $sp^3d^2$, octahedral (4) $dsp^2$, tetrahedral

**78.** Given below are the two statements
**Statement I:** $SO_2$ is a colourless gas and is highly soluble in water.
**Statement II:** $SO_2$ reacts with chlorine in the presence of charcoal to give thionyl chloride.
In light of above statements, choose the correct answer.
(1) Statement I is correct but statement II is incorrect.
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct.
(4) Both statement I and statement II are incorrect.

**79.** What is the correct electronic configuration of the central atom in $[Co(H_2O)_6]^{3+}$ based on crystal field theory?
(1) $e^2t_2^4$ (2) $t_{2g}^6e_g^0$
(3) $t_{2g}^4e_g^0$ (4) $e^3t_2^3$

**80.** Given below are two statements :
**Statement I :** $[Co(NH_3)_6]^{3+}$ is a homoleptic complex whereas $[Co(NH_3)_4Cl_2]^+$ is a heteroleptic complex
**Statement II:** cis-$[CrCl_2(ox)_2]^{3-}$ is a optically active complex ion whereas trans-$[CrCl_2(ox)_2]^{3-}$ is a optically inactive complex ion.
In the light of the above statements, choose the correct answer from the option given below :
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

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**81.** The hybridisation of atomic orbitals of Xe in $XeO_3$, $XeOF_4$ and $XeO_2F_2$ are

(1) $sp^3$, $sp^3d^2$, $sp^3$ respectively

(2) $sp^3$, $sp^3d^2$, $sp^3d$ respectively

(3) $sp^3d$, $sp^3d^2$, $sp^3$ respectively

(4) $sp^3$, $sp^3d$, $sp^3d^2$ respectively

**82.** In the following question, a statement of assertion (A) is followed by a statement of reason (R)

**Assertion (A):** Pyridine is more basic than aniline.

**Reason (R):** Pyridine is a heterocyclic compound.

In the light of above statement, choose the correct answer.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

**83.** Given below are two statements.

**Statement I :** β-black phosphorous is prepared by heating white phosphorous at 473 K under high pressure.

**Statement II :** Red phosphorous is obtained by heating white phosphorous at 573 K in an inert atmosphere.

In the light of above statements choose the correct option.

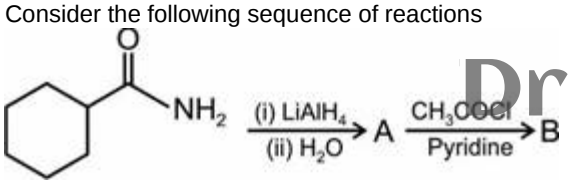
(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

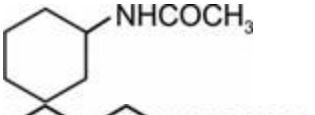
(3) Both statement I and II are correct

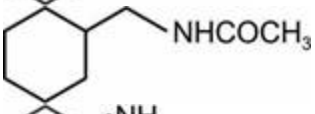
(4) Both statement I and II are incorrect

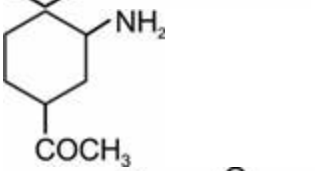
**84.** Consider the following sequence of reactions

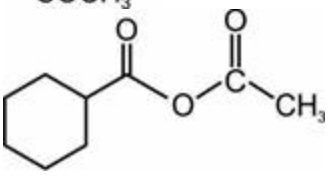



Product B is

(1) 


(2) 


(3) 


(4) 


**85.** Which among the following is not a group-IV cation?

(1) $Ni^{2+}$ (2) $Zn^{2+}$

(3) $Pb^{2+}$ (4) $Co^{2+}$

**86.** Approximate percentage of lanthanoids in mischmetall is

(1) 5% (2) 75%

(3) 25% (4) 95%

**87.** Permanganate titrations in presence of hydrochloric acid are unsatisfactory since

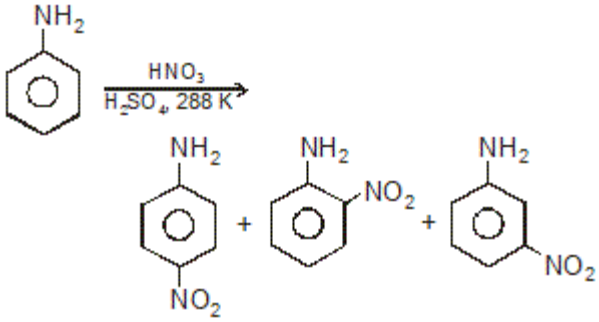
(1) Hydrochloric acid is oxidised to hydrogen

(2) Hydrochloric acid is oxidised to chlorine

(3) Hydrochloric acid is highly volatile in nature

(4) Hydrochloric acid forms salt with permanganate ion

**88.** For the reaction




The correct increasing order of the amount of products formed is

(1) Ortho < Para < Meta (2) Meta < Ortho < Para

(3) Meta < Para < Ortho (4) Ortho < Meta < Para

**89.** Match the complexes in **List-I** with their spin only magnetic moment (in BM) in **List-II**.

| **List-I** | **List-II** |
|---|---|
| a. $[Co(H_2O)_6]^{3+}$ | (i) $\sqrt{24}$ |
| b. $[Fe(CN)_6]^{3-}$ | (ii) $\sqrt{35}$ |
| c. $[Fe(H_2O)_6]^{2+}$ | (iii) 0 |
| d. $[Mn(H_2O)_6]^{2+}$ | (iv) $\sqrt{3}$ |

Choose the correct option

(1) a(i), b(ii), c(iii), d(iv) (2) a(iv), b(iii), c(ii), d(i)

(3) a(iii), b(iv), c(i), d(ii) (4) a(ii), b(iii), c(iv), d(i)

**90.** The correct name for the complex $[CrCl_2(H_2O)_4]NO_3$ is

(1) Dichlorotetraaqueouschromium (III) nitrate

(2) Tetraaquadichlorochromate (III) nitrate

(3) Dichlorotetraaqueouschromium (IV) nitrate

(4) Tetraaquadichlorochromium (III) nitrate

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**91.** Which of the following rRNA is not synthesized by RNA polymerase-I ?

(1) 5 S rRNA (2) 28 S rRNA

(3) 18 S rRNA (4) 5.8 S rRNA

**92.** Which of the following enzymes is responsible for the hydrolysis of lactose in the bacterial cell?

(1) Transacetylase (2) Permease

(3) β-galactosidase (4) Polymerase

**93.** The last step of DNA fingerprinting is

(1) Blotting (2) Hybridisation

(3) Electrophoresis (4) Autoradiography

**94.** How many amino acids will be coded by the sequence if the $12^{th}$ base of given mRNA converts to 'G'?

5′AUGGUGUGAUACACC3′

(1) Two (2) Three

(3) Five (4) Four

**95.** Which of the following nitrogenous bases are common for both RNA and DNA?
(1) C, G, A (2) G, A, U
(3) T, A, C (4) U, A, C

**96.** If *E.coli* is growing in a medium that has lactose, then
(1) Repressor will bind to inducer
(2) There will be no synthesis of repressor protein
(3) There will be no synthesis of enzymes of structural genes
(4) Repressor will bind to operator

**97.** Bond formed between two adjacent nucleotides of a DNA strand is
(1) Glycosidic linkage (2) Peptide bond
(3) Phosphodiester bond (4) H-bond

**98.** Read the following statements (A-D)
(A) The promoter and terminator flank the structural gene in a transcription unit
(B) The structural gene in a transcription unit is monocistronic mostly in bacteria.
(C) Exons are intervening sequences that do not appear in mature or processed RNA.
(D) The transcription and translation can be coupled in bacteria.
The **correct** ones are
(1) Only (A) and (B) (2) Only (A), (B) and (D)
(3) Only (B) and (D) (4) Only (A) and (D)

**99.** Mark the **incorrect** statement w.r.t. features of human genome.
(1) The human genome contains 3164.7 million bp
(2) The average gene consists of 3000 bases
(3) Less than 2 percent of genome codes for proteins
(4) Chromosome 1 has fewest genes

**100.** Amongst the following, which one has least number of nucleotides in its genome?
(1) $\phi \times 174$ bacteriophage (2) *E. coli*
(3) Human (4) Lambda phage

**101.** Read the following statements and choose the **correct** option.
**Statement A:** The development and differentiation of embryo into adult organisms are also a result of the coordinated regulation of expression of several sets of genes.
**Statement B:** James Watson postulated the presence of an adaptor molecule for translation.
(1) Only statement B is correct
(2) Only statement A is correct
(3) Both the statements A and B are correct
(4) Both the statements A and B are incorrect

**102.** Which of the given is one of the hallmarks of DNA double helix model of Watson and Crick?
(1) It is made up of two polynucleotide chains forming backbone of DNA
(2) The two chains run parallel in polarity
(3) The base pairing between two strands of polynucleotide chains
(4) The two chains are coiled in a left handed fashion only

**103.** Arrange the following steps that occur during treatment of municipal waste water.
a. Soil and small pebbles are removed by sedimentation.
b. Aerobic microbes consume the major part of organic matter in the effluent.
c. Anaerobic bacteria digest the aerobic bacteria & fungi in the sludge
d. Bacterial flocs are allowed to sediment in settling tank
e. Removal of floating debris by sequential filtration
(1) a, e, d, b, c (2) e, a, d, b, c
(3) a, e, b, c, d (4) e, a, b, d, c

**104.** Select **incorrect** match.
(1) Citric acid – *Aspergillus niger*
(2) Streptokinase – *Streptococcus*
(3) Butyric acid – *Clostridium*
(4) Cyclosporin A – *Monascus purpureus*

**105.** Statin acts by competitively inhibiting the enzyme responsible for synthesis of
(1) Citric acid (2) Lactic acid
(3) Acetic acid (4) Blood cholesterol

**106.** Biocontrol agents are preferred over chemical methods of pests and disease control because
a. They are non toxic for non targeted organisms
b. They do not kill useful organisms
c. They keep pests at manageable levels
(1) Only a (2) Only b
(3) Only b and c (4) All a, b and c

**107.** Organism responsible for the alcohol production during fermentation is
(1) A unicellular bacterium (2) A multicellular fungus
(3) A unicellular fungus (4) A protist

**108.** Swiss cheese is ripened by
(1) *Penicillium roquefortii*
(2) *Penicillium camembertii*
(3) *Propionibacterium sharmanii*
(4) *Penicillium notatum*

**109.** Fermentors are used for growing microbes to
(1) Synthesise chemical pesticides
(2) Produce biofertilisers
(3) Increase BOD of sewage waste
(4) Synthesising industrial products

**110.** Which of the given statements is/are **true** w.r.t. antibiotics?
a. The first antibiotic streptomycin was discovered by Alexander Fleming.
b. Antibiotics have greatly improved our capacity to treat deadly diseases.
c. Fleming, Chain and Florey were awarded the Nobel prize in 1945 for discovery of antibiotic.
(1) a only (2) b and c only
(3) a and c only (4) All a, b and c

**111.** Which of the following is **not** true with respect to LAB?
(1) Acids produced by LAB completely digest the milk proteins
(2) A small amount of curd contains millions of LAB
(3) They multiply at suitable temperatures
(4) They convert milk to curd, increasing its nutritional quality

**112.** An association in which one species is harmed and the other is not affected is known as
(1) Commensalism (2) Amensalism
(3) Brood parasitism (4) Mutualism

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**113.** The integral form of exponential growth equation will be
(1) $N_t - N_0 = e^{rt}$ (2) $N_t = N_0 e^{rt}$
(3) $\frac{N_t}{N_0} = \frac{1}{e^{rt}}$ (4) $N_t \times N_0 = e^{rt}$

**114.** Find the **odd** one w.r.t. mutualism
(1) Lichens (2) Mycorrhizae
(3) Fig and fig wasp (4) Barnacles on whale

**115.** Mark True (T) or False (F) for the given statements.
**Statement I :** The Mediterranean orchid *Ophrys* employs 'sexual deceit' to get pollination done by a species of bee.
**Statement II :** Pacific salmon fish and Bamboo breed only once in their lifetime.
(1) **I** – T, **II** – F (2) **I** – F, **II** – T
(3) **I** – T, **II** – T (4) **I** – F, **II** – F

**116.** If in a pond there were 50 fishes last year and through reproduction, 10 new fishes are added, taking the current population to 60. What is the birth rate per fish per year?
(1) 0.2 (2) 0.4
(3) 0.1 (4) 0.25

**117.** Read the following statements and select the **correct** option regarding age-pyramid.
(i) An age pyramid is a graphic representation of proportion of various age groups of a population.
(ii) The shape of the pyramid reflects the growth status of the population.
(iii) In bell shaped age-pyramid, population is said to be stable.
(iv) Urn-shaped age-pyramid shows positive growth.
(1) (i), (ii) and (iv) are correct (2) Only (iii) is correct
(3) All are correct except (ii) (4) All are correct except (iv)

**118.** Read the given statements and select the **correct** option.
**Statement-A :** Prey species have evolved various defences to minimize the impact of predation.
**Statement-B :** Life cycle of endoparasites is more complex due to their extreme specialization.
(1) Only statement-A is correct
(2) Only statement-B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

**119.** Read the following statements and choose the correct option.
**Assertion :** Cattle never grazes on *Calotropis*.
**Reason :** *Calotropis* contains a poisonous chemical, called cardiac glycosides.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

**120.** Which of the following are **not** given any place in ecological pyramids?
(1) Primary producers (2) Primary consumers
(3) Top carnivores (4) Saprophytes

**121.** Read the following statements
**Statement (A):** Both eukaryotic and prokaryotic organisms are involved in the degradation of detritus into simpler inorganic substance.
**Statement (B):** Dead remains of the animals and dead remains of plants such as fallen leaves, bark and flowers are the raw material for decomposition.
In the light of the above statements, choose the **correct** answer from the options given below.
(1) Only statement (A) is correct
(2) Only statement (B) is correct
(3) Both the statements (A) and (B) are correct
(4) Both the statements (A) and (B) are incorrect

**122.** Select the **odd** one w.r.t decomposition
(1) Detritivores break down detritus into smaller particles by a process called fragmentation
(2) By the process of leaching, organic substances go down to soil horizon
(3) Bacterial and fungal enzymes are important for degradation of detritus
(4) Decomposition is largely an oxygen requiring process

**123.** Read the following statements and select the **correct** option.
A. Fungi and bacteria can be decomposers.
B. Decomposition is slow at low temperature and reduced aeration.
(1) Only A is correct
(2) Only B is correct
(3) Both A and B are correct
(4) Both A and B are incorrect

**124.** Select the **incorrect** statement
(1) Energy flow in an ecosystem follows the second law of thermodynamics
(2) GFC is the major conduit for energy flow in an aquatic ecosystem
(3) In a terrestrial ecosystem, much larger fraction of energy flows through detritus food chain than GFC
(4) Detritus food chain begins with different detrivores, further consumed by consumers

**125.** Which of the following ecological pyramids is **not** possible for a tree ecosystem?
(1) Inverted pyramid of number
(2) Upright pyramid of energy
(3) Spindle-shaped pyramid of number
(4) Inverted pyramid of biomass

**126.** What is the percentage of PAR (Photosynthetically active radiation) in incident solar radiation?
(1) < 50% (2) > 50%
(3) 50% (4) 0%

**127.** Choose the **odd** one w.r.t. humus.
(1) Dark coloured amorphous substance
(2) Colloidal in nature
(3) Undergoes decomposition at very fast rate
(4) Serves as a reservoir of nutrients

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**128.** Match list I with list II and choose the **correct** option.

| | List-I | | List-II |
|---|---|---|---|
| A. | Secondary productivity | (i) | It is an available biomass for the consumption to heterotrophs |
| B. | Primary production | (ii) | It is the sum of NPP and respiratory losses |
| C. | Net primary productivity | (iii) | The amount of biomass produced per unit-area over a time period by plants during photosynthesis |
| D. | Gross primary productivity | (iv) | Rate of formation of new organic matter by consumers |

(1) A-(iii), B-(iv), C-(ii), D-(i) (2) A-(iv), B-(iii), C-(i), D-(ii)
(3) A-(iii), B-(i), C-(ii), D-(iv) (4) A-(iv), B-(i), C-(ii), D-(iii)

**129.** Read the following statements and select the **correct** option

**Statement A:** Colombia located near the equator has nearly 1,400 species of birds while New York at 41° N has 105 species.

**Statement B:** India, with much of its land area in the tropical latitudes, has more than 1,200 species of birds.

(1) Both the statements A and B are incorrect
(2) Only statement A is correct
(3) Only statement B is correct
(4) Both the statements A and B are correct

**130.** When a host fish species becomes extinct, its unique assemblage of parasites also meets the same fate. It is an example of

(1) Co-extinction (2) Alien species invasion
(3) Over-exploitation (4) Habitat loss

**131.** Find the **odd** statement w.r.t. biodiversity hotspots.

(1) Very high levels of species richness
(2) Species found in these regions show very low degree of endemism
(3) All biodiversity hotspots put together cover less than 2 percent of Earth's land mass
(4) Strict protection of these hotspots will reduce the ongoing mass-extinction by almost thirty percent

**132.** Which of the following is **not** an *ex-situ* conservation strategy?

(1) Biodiversity hotspot (2) Wildlife safari park
(3) Seed bank (4) Botanical garden

**133.** In an ecosystem, rich species diversity is important to provide all of the following, **except**

(1) Resistance to invasions by alien species
(2) Resistance to occasional natural disturbances
(3) Resilience to man-made disturbances
(4) Much variations in productivity from year to year

**134.** Match the columns and select the **correct** option.

| | Column A (Extinct species) | | Column B (Native area) |
|---|---|---|---|
| a. | Dodo | (i) | Russia |
| b. | Quagga | (ii) | Mauritius |
| c. | Thylacine | (iii) | Africa |
| d. | Steller's sea cow | (iv) | Australia |

| | a | b | c | d |
|---|---|---|---|---|
| (1) | (ii) | (iv) | (iii) | (i) |
| (2) | (i) | (ii) | (iii) | (iv) |
| (3) | (ii) | (iii) | (iv) | (i) |
| (4) | (iv) | (iii) | (ii) | (i) |

(1) (1) (2) (2)
(3) (3) (4) (4)

**135.** Read the following Assertion (A) and Reason (R) and choose the **correct** option.

**Assertion (A):** For many taxonomic groups, species inventories are more complete in tropics than in temperate regions.

**Reason (R):** As compared to the tropical regions, large proportion of the species waiting to be discovered are in the temperate regions.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) Both (A) and (R) are false

## ZOOLOGY

**136.** Which of the following is an **incorrect** statement?

(1) Most parasites are pathogens as they cause harm to host by living in or on them
(2) Widal test helps to diagnose the STD named Syphilis
(3) Pneumonia is caused by bacterium *Streptococcus pneumoniae*
(4) Dysentery and plague are bacterial diseases seen in humans

**137.** How many of the organs mentioned in the box below are involved in differentiation of immature lymphocytes into antigen-sensitive lymphocytes in humans?

Bone marrow, Spleen, Tonsils, Thymus, Lymph nodes

Choose the **correct** option.

(1) Three (2) Four
(3) Two (4) Five

**138.** The most serious and possibly fatal malignant malaria causing species of *Plasmodium* is

(1) *vivax* (2) *malaria*
(3) *falciparum* (4) *ovale*

**139.** Which drug can be used for treatment of insomnia?

(1) Cocaine (2) Barbiturates
(3) Crack (4) Caffeine

**140.** The most feared property of malignant tumours is

(1) Remaining confined to their original location
(2) Encapsulation
(3) Metastasis
(4) Lysis of RBCs

**141.** In the box given below, some barriers of innate immunity are mentioned.

PMNL, NK cell, HCl in stomach, Skin, Saliva, Mucus coating of urinogenital tract, Interferons, Tears

How many of them are categorised under the physiological barriers?

(1) Two (2) Three
(3) Four (4) Five

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**142.** Read the following statements and choose the option with only correct statements.
a. Dengue and chikungunya are vector-borne viral diseases.
b. Biotechnology is on the verge of making newer and safer vaccines available to public.
c. Physiological barriers include mucus coating of the epithelium lining the respiratory, urogenital and gastrointestinal tracts.
d. Skin is the main physical barrier which prevents entry of micro-organisms.
e. Acquired immunity provides different types of barriers to prevent the entry of foreign agents into our body.
(1) a, b, c (2) b, c, d
(3) a, b, d (4) Only c and e

**143.** **False** statement w.r.t. secondary immune response is that it is
(1) Mediated through IgG
(2) Slow and short lived response
(3) An immune response triggered by memory cells
(4) Also termed as anamnestic response

**144.** **Assertion (A):** Amphetamines and steroids are often abused by sportspersons.
**Reason (R):** Amphetamines are strong stimulants that help fight fatigue and induce euphoria while steroids produce anabolic effects.
In the light of above statements choose the correct option.
(1) Both Assertion and Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion and Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

**145.** Select the disease which spreads *via* houseflies.
(1) Amoebiasis (2) Malaria
(3) Dengue (4) Filariasis

**146.** Read the features given below w.r.t. common cold.
A. Caused by Rhino viruses
B. Affects the complete respiratory passage including lungs
C. Usually lasts for 3-7 days
D. Non-infectious human aliment
Select the option including only correct features
(1) A and B (2) A and C
(3) A, B and C (4) A, B, C and D

**147.** The two key concepts of Darwinian theory of evolution are
(1) Natural selection and branching descent
(2) Natural selection and mutation
(3) Mutation and branching descent
(4) Embryology and mutation

**148.** Match the Column A and B and choose the correct option w.r.t. human ancestors and their cranial capacities

| | Column A | | Column B |
|---|---|---|---|
| (i) | Neanderthal man | a. | 1650 cc |
| (ii) | *Homo erectus* | b. | 650 - 800 cc |
| (iii) | *Homo habilis* | c. | 1400 cc |
| (iv) | Cro-magnon man | d. | 900 cc |

(1) i-b, ii-a, iii-d, iv-c (2) i-c, ii-a, iii-b, iv-d
(3) i-d, ii-a, iii-c, iv-b (4) i-c, ii-d, iii-b, iv-a

**149.** The fundamental components of early atmosphere were most likely
(1) $NH_3$, $CH_4$, $H_2$ and $H_2O$ vapour
(2) $H_2$, $O_2$, $CH_4$ and $H_2O$ vapour
(3) $CH_4$, $O_3$, $N_2$ and $H_2O$ vapour
(4) $H_2O$, $CH_4$ and $NH_3$ only

**150.** Among the following, select the structures that are result of convergent evolution
(1) Heart of whale and man
(2) Eye of *Octopus* and bat
(3) Forelimbs of cheetah and bat
(4) Brain of bat and man

**151.** Naturalist who worked in Malay Archipelago and come to similar conclusions around the theory of natural selection was
(1) Charles Darwin (2) Lamarck
(3) Alfred Wallace (4) Thomas Malthus

**152.** Select the **incorrect** statement w.r.t industrial melanism.
(1) Before industrialisation set in, there were more white-winged moths than dark-winged moths on trees.
(2) It is an example of evolution caused due to anthropogenic actions.
(3) Post-industrialisation, white-winged moths survived and melanised moths did not survive due to predators.
(4) No variant got completely wiped out.

**153.** Genetic variability has decreased considerably in the Cheetah population in Africa after hunting over the course of time making the present day Cheetah population susceptible to a number of fatal diseases. This is a classic example of
(1) Migration (2) Gene flow
(3) Bottle neck effect (4) Founder effect

**154.** Read the given statements
(a) Crossing over during fusion of gametes results in formation of new combination of characters.
(b) Natural selection can lead to change in allelic frequencies.
(c) According to Lamarck, use and disuse of organs is major driving force behind evolution of life forms.
(d) Increase in number of heterozygotes [$Hb^A/Hb^S$] over homozygotes [$Hb^A/Hb^A$ or $Hb^S/Hb^S$] in malaria prone area is an example of directional selection.
Select the option that represents only incorrect statement
(1) (a) and (c) (2) (b) and (d)
(3) (a) and (d) (4) (c) and (d)

**155.** If the frequency of occurrence of dominant allele 'A' is p and that of recessive allele 'a' is q. Then, according to Hardy-Weinberg law 'q' is equal to.
(1) 1 + p (2) $1 - p^2 - 2pq$
(3) 1 – p (4) $1 - q^2$

**156.** What kind of natural selection operates, when nature selects only one extreme trait of a character?
(1) Stabilising selection (2) Disruptive selection
(3) Directional selection (4) Balancing selection

**157.** 'They' used hides to protect their body and buried their dead. 'They' refers to
(1) *Dryopithecus* (2) *Ramapithecus*
(3) *Australopithecus* (4) Neanderthal man

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**158.** Comprehend the **given** statements.
(a) The first organisms that invaded land were plants.
(b) Bryophytes are the direct descendants of Psilophyton.
(c) About 85 mya, the dinosaurs suddenly disappeared from the Earth.
(d) The first mammals were like shrews.
How many of the above given statements is/are incorrect?
(1) One (2) Two
(3) Three (4) Four

**159.** Match the column I with column II and select the **correct** option.

| | Column I | | Column II |
|---|---|---|---|
| a. | Biolistics | (i) | Also called molecular glue |
| b. | DNA ligases | (ii) | Used as vector |
| c. | Bacteriophages | (iii) | Method of vectorless gene transfer |
| | | (iv) | Pathogenic to dicots |

(1) a(iii); b(i); c(ii) (2) a(ii); b(iii); c(i)
(3) a(iii); b(iv); c(i) (4) a(iv); b(i); c(ii)

**160.** If a foreign DNA is cut with an enzyme 'X' that recognizes the palindromic sequence 5' GAATTC 3'. Then, for successful ligation of foreign DNA with the cloning vector, the vector should be cleaved using
(1) *Bam*H I (2) *Eco*R I
(3) *Pst* I (4) *Sal* I

**161.** Enzyme used in PCR for amplification of particular DNA fragment is _____. It is of _____ nature and is obtained from _____.
Choose the option that fill the respective blanks **correctly**.
(1) *Taq* polymerase, Thermostable, *Thermus aquaticus*
(2) *Taq* polymerase, Thermolabile, *Thermus aquaticus*
(3) DNA polymerase I, Thermophilic, *Trichophyton*
(4) DNA polymerase II, Thermophilic, *Thermococcus*

**162.** Agarose is a natural polymer extracted from seaweed. It is used in
(1) Polymerase Chain Reaction
(2) ELISA
(3) Gel electrophoresis
(4) Downstream processing

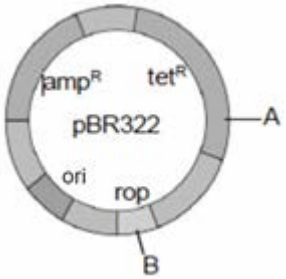
**163.** Downstream processing includes
(1) Expression of protein in heterologous host
(2) Purification of DNA from cells
(3) Purification and preservation of desired product after biosynthesis
(4) Large scale culture of recombinant bacteria in bioreactors

**164.** During blue/white selection, among the colonies of transformants, recombinants and non-recombinants respectively produce
(1) Blue and white colured colonies
(2) White coloured colonies only
(3) White and blue coloured colonies
(4) Blue coloured colonies only

**165.** The first clinical gene therapy in 1990 was given for treating
(1) Myasthenia gravis
(2) Small pox
(3) Kidney stone
(4) Adenosine deaminase deficiency

**166.** Read statements A and B and choose the **correct** option.
**Statement-A:** Insulin cannot be orally administered to diabetic patient because it gets degraded in alimentary canal.
**Statement-B:** The two polypeptides of human insulin are linked together by only hydrogen bonds.
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Statement A is correct but statement B is incorrect
(4) Statement A is incorrect but statement B is correct

**167.** Identify A and B restriction site in below given figure of plasmid.




(1) *Pvu* I and *Pst* I respectively
(2) *Hind* III and *Pvu* I respectively
(3) *Sal* I and *Pvu* II respectively
(4) *Bam* HI and *Cla* I respectively

**168.** The construction of first recombinant DNA was accomplished by linking a foreign gene with native plasmid of the bacterium
(1) *Salmonella typhimurium* (2) *E. coli*
(3) *Agrobacterium* (4) *Bacillus thuringiensis*

**169.** A piece of DNA is subjected to restriction enzyme digestion and the digested sample is electrophoresed. Which statement is **correct** in the above context?
(1) Smallest fragment is present near cathode
(2) Largest fragment is present near anode
(3) Largest fragment is present in the middle of gel
(4) Smallest fragment is present near anode

**170.** Which of the following organisms is **not** correctly matched with its cell wall degrading enzyme?
(1) Plant cells – Cellulase (2) Algae – Methylase
(3) Fungi – Chitinase (4) Bacteria – Lysozyme

**171.** A technique in which, recombinant DNA is directly injected into the nucleus of an animal is
(1) Electroporation (2) Microinjection
(3) Biolistics (4) Downstreaming

**172.** If any protein encoding gene is expressed in a heterologous host then it is called as
(1) Recombinant protein (2) Derived protein
(3) Refined protein (4) First class protein

**173.** A recombinant DNA in which the desired gene sequence is inserted in the coding sequence of enzyme β-galactosidase is used for mass scale production of a product in bacterial host cells.
The host bacteria having the recombinant DNA in them can be easily selected because
(1) They have ability to produce colour in presence of chromogenic substrate, 'X-gal'
(2) They are unable to produce colour due to lack of ability to utilize 'X-gal'
(3) They have insertionally inactivated antibiotic resistance genes
(4) They have active β-galactosidase gene

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**174.** If a circular plasmid contains 4 sites for restriction enzyme *Pst* I, how many fragments of DNA will be obtained upon restriction digestion by given enzyme?
(1) 4 (2) 8
(3) 3 (4) 5

**175.** Maintenance of sterile *i.e.*, microbial contamination- free, ambience in chemical engineering processes is important to
(1) Enable growth of only desired microbes
(2) Enable growth of all microbes
(3) Enable growth of harmful microbes
(4) Disable growth of desired microbes

**176.** α-1-antitrypsin is used in the treatment of
(1) Myocardial infarction (2) Marasmus
(3) Emphysema (4) Asthma

**177.** In the following question, a statement of assertion (A) is followed by a statement of reason (R).
**Assertion (A):** Humulin is synthesized usually in heterologous host *E.coli*.
**Reason (R):** *Escherichia coli* lacks a native plasmid.
In the light of above statements, select the **correct** option.
(1) Both Assertion and Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion and Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

**178.** Which of the following transgenic animals are being used to test the safety of the polio vaccine?
(1) Rabbits (2) Pigs
(3) Mice (4) Sheep

**179.** In India how many recombinant therapeutics are being presently marketed?
(1) 30 (2) 20
(3) 12 (4) 21

**180.** How many of the following given statements is/are **correct**?
I. GEAC makes decisions only regarding patents of various new discoveries in India.
II. About 95% of all existing transgenic animals are mice.
III. Among 2000 varieties of rice in India, Basmati variety was patented by a US company
IV. RNAi is a post translational method of gene silencing.
Choose the **correct** option
(1) Four (2) Three
(3) Two (4) One

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